

CHENNAI INSTITUTE OF TECHNOLOGY (AUTONOMOUS)

Sarathy Nagar, Kundrathur, Chennai-600069

Approved by AICTE and Affiliated to Anna University, Chennai

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

INTERNSHIP



A Report on Internship
By

NAME: ILAM BHARATHI M
REG.NO: 24EC0066

ECE-B

NOV-DECEMBER 2025

Vision of the Institute:

To be an eminent centre for Academia, Industry and Research by imparting knowledge, relevant practices and inculcating human values to address global challenges through novelty and sustainability.

Mission of the Institute:

- IM1** To create next generation leader by effective teaching learning methodologies and instill scientific spark in them to meet the global challenges.
- IM2** To transform lives through deployment of emerging technology, novelty and sustainability.
- IM3** To inculcate human values and ethical principles to cater the societal needs.
- IM4** To contribute towards the research ecosystem by providing a suitable, effective platform for interaction between industry, academia and R &D establishments.

To Excel in the emerging areas of Electronics and Communication Engineering by imparting knowledge, relevant practices and inculcating human values to transform the students as potential resources to cater the industrial and societal development through sustainable technology growth.

Mission of the Department:

DM1: To provide strong fundamentals and technical skills through effective teaching learning Methodologies.

DM2: To transform lives of the students by fostering ethical values, creativity and innovation to become Entrepreneurs and establish Start-ups.

DM3: To habituate the students to focus on sustainable solutions to improve the quality of life and welfare of the society.

DM4: To provide an ambience for research through collaborations with industry and academia.

DM5: To inculcate learning of emerging technologies for pursuing higher studies leading to lifelong learning.

CHENNAI INSTITUTE OF TECHNOLOGY

An Autonomous Institute

CHENNAI-69



BONAFIDE CERTIFICATE

This is to certify that the “**Internship Report**” Submitted by **ILAM BHARATHI M**
(**Regno: 24EC0066**) is the work done by him and submitted during the academic year
2025-2026, in partial fulfilment of the requirements for the award of the degree
of **BACHELOR OF ENGINEERING** in **DEPARTMENT OF ELECTRONICS**
AND COMMUNICATION ENGINEERING
at **COE-VLSI(Cadence Lab)**.

Department Internship Coordinator

Head of the Department

Internal Examiner
Review Evaluation/Comments

External Examiner Internship

Sl No.	Criterion	Max. Marks	Marks Allotted
1.	Regularity in maintenance of the diary.	10	
2.	Adequacy & quality of information recorded	10	
3.	Drawings, sketches and data recorded	10	
4.	Thought process and recording techniques used	05	
5.	Organization of the information	05	
6.	Originality of the Internship Report	10	
7.	Adequacy and purposeful write-up of the Internship Report	10	
8.	Organization, format, drawings, sketches, style, language etc. of the Internship Report	10	
9.	Practical applications, relationships with basic theory and concepts	10	
10.	Presentation Skills	20	
Total		100	

External Internship Advisor's Name:

Company:

Date:

Signature

ACKNOWLEDGEMENT

First and foremost, I humbly express our gratitude to the Almighty for His unending grace in all our undertakings. It is a privilege to thank the individuals who made this project a reality and provided their encouragement, guidance, and support throughout the entire process of completing my project at Chennai Institute of Technology, Kandrathur.

I wish to extend our heartfelt thanks to our respected Chairman, **Mr. P. SRIRAM**, for his significant support and motivation in promoting technological advancements and pioneering new projects.

I sincerely thanks to Prof. **Dr. A. RAMESH, B.E., M.E., Ph.D.**, Principal of Chennai Institute of Technology, for his insightful guidance, steadfast support, and encouragement in the successful completion of this project.

I express my gratitude to Prof. **Dr. T S.Arulananth Ph.D.**, Head of Department, Department of Electronics and Communication Engineering, Chennai Institute of Technology, for guiding and encouraging during the project.

I thank my supervisor, **MRS.S.Aathilakshmi M.E.**, Assistant Professor at Chennai Institute of Technology, for his invaluable guidance and support from the project's start to its final edit.

I'm grateful for the kindness and support of our classmates, whose patience and help were invaluable, especially in formatting this project. I also sincerely thank my family for their constructive support.

PREFACE

I am a student of the **Department of Electronics and Communication Engineering** at **Chennai Institute of Technology**, and this project has been carried out as part of my academic curriculum under the **Chip to Startup Initiative**. The main objective of this project is to strengthen my practical understanding by applying the theoretical concepts learned during my course of study to a real-time VLSI design problem.

This project provided a valuable opportunity to correlate classroom learning with hands-on design and analysis. Through this work, I gained practical exposure to circuit-level design, simulation, and performance evaluation, which enhanced my understanding of **CMOS-based system design and digital-analog interfacing techniques**.

Although this report may reflect certain limitations due to my academic learning phase, it represents a sincere and structured effort to analyze and implement the proposed system effectively. The outcomes of the project are presented in an organized manner, detailing the design methodology, operational principles, and performance analysis of the system.

This report has been prepared as part of my **Second-Year academic project**, and it reflects my interest in innovation and applied research in the field of electronics. As an engineering student, I believe it is my responsibility to utilize my technical knowledge to develop solutions that address real-world challenges. With this perspective, the project was carried out under the **Chip to Startup ecosystem**, which promotes practical learning, research orientation, and innovation in semiconductor and VLSI technologies.

TABLE OF CONTENTS

SL. NO.	TITLE	PAGE NO.
1	Abstract	20
2	Introduction	21
3	Project Objective	22
3.1	Significance of the Study	22
3.2	Applications	22
4	Literature Survey	23
5	Limitation of the Existing System	24
5.1	Summary of the Research Findings	24
6	Research Methodology	25
7	Proposed Methodology	26
8	Results and Discussion	34
9	Comparison with Existing CDC Designs	36
10	Conclusion	40

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

1st Week	Date	Day	Work done
	03.11.2025	Monday	Studied the fundamentals of capacitance-to-digital converters and understood the overall CDC system architecture.
	04.11.2025	Tuesday	Analysed frequency-domain CDC architectures and identified the role of major functional blocks
	05.11.2025	Wednesday	Studied switched-capacitor feedback techniques for improving linearity and stability in CDC systems.
	06.11.2025	Thursday	Learned the operating principles of ring oscillators and their frequency control mechanisms.
	07.11.2025	Friday	Conducted a literature survey on existing CDC designs and identified key limitations.

2nd Week	Date	Day	Work done
	10.11.2025	Monday	Designed the capacitive sensor interface and switched-capacitor network using CMOS principles

	11.11.2025	Tuesday	Designed and analysed the OTA used for biasing and frequency control of the ring oscillator
	12.11.2025	Wednesday	Implemented and simulated the ring oscillator in Cadence Virtuoso using 180 nm CMOS technology
	13.11.2025	Thursday	Designed a D flip-flop-based divide-by-2 frequency divider to obtain a 50% duty-cycle output
	14.11.2025	Friday	Verified the frequency divider functionality through simulation and waveform analysis

3rd Week	Date	Day	Work done
	17.11.2025	Monday	Integrated all CDC sub-blocks including sensor interface, OTA, ring oscillator, and frequency divider.
	18.11.2025	Tuesday	Performed timing and sampling analysis to evaluate the effect of divider ratio on CDC performance.
	19.11.2025	Wednesday	Analysed frequency-to-digital conversion behaviour for varying input capacitance values.
	20.11.2025	Thursday	Evaluated CDC performance in terms of resolution, frequency range, and power consumption.

	21.11.2025	Friday	Compared the proposed CDC with existing designs reported in literature.
--	-------------------	---------------	---

4th Week	Date	Day	Work done
	22.11.2025	Monday	Implemented full-custom CDC circuits in 180 nm CMOS using Cadence Virtuoso
	23.11.2025	Tuesday	Performed simulation-based verification and functional validation of the CDC system
	24.11.2025	Wednesday	Analysed the impact of parasitic, noise, and duty-cycle distortion on CDC operation.
	25.11.2025	Thursday	Optimized circuit parameters to improve linearity, duty-cycle accuracy, and power efficiency
	26.11.2025	Friday	Compiled results and prepared the final CDC project report for review evaluation

ABSTRACT

The manuscript now reports on a time-encoded CDC designed specifically for microfluidic lab-on-a-chip systems using IDEs. The reported approach directly connects to the low capacitance values measured by the CDC, eliminating the need for conventional constant charge biasing, which typically requires charge pumps and high value resistors in charge-sensitive CDCs. The presence of a SC feedback loop in the time-encoded CDC using an RO-loop provides seamless on-chip realization in a bias-free environment. The microfluidic channel permittivity variations are resolved into corresponding oscillation frequencies to provide precise digitization of capacitance values in the vicinity of a nominal capacitance value of 2.575 pF. A prototype designed in 180-nm CMOS technology has an active area of 0.025 mm² and a power consumption of 1343 μ W. Also, measurement results indicate capacitance noise resolution in the attofarad-level between 1.7 aF to 2.9 aF RMS in a signal bandwidth ranging between 300 Hz and 6.8 kHz, surpassing the resolution achieved in conventional CDC structures and comparable to that achieved in commercial capacitance measurement instruments. Due to the characteristics of high sensitivity, small size, low power, and bias-free detection, the proposed CDC is particularly suited to next-generation microfluidic systems, such as label-free bioanalytical analysis and single entity detection.

INTRODUCTION

Many microfluid operations such as droplet detection, flow monitoring, concentration measurement and fluidic composition analysis - fundamentally rely on tracking variations in the dielectric properties or volumetric displacement of the fluid inside a microchannel. In this work we propose modified SC-feedback. CDC tailored specifically for microfluidic applications Successive approximation register (SAR) based capacitance-to-digital converters are well known for their excellent energy efficiency, but linearity and resolution are limited, and clock frequency, Time-encoded CDCs convert capacitance changes into frequency, which can then be easily processed by digital logic. Switched-capacitor behaviour depends on capacitance ratios and clock frequency not on absolute resistor values, it has a predictable behaviour with good matching. A good approach for capacitance-to-digital converters is to remove the intermediate voltage conversion and to directly encode the capacitive signal into the oscillation frequency of a ring oscillator. This approach eliminates the usage of large charge pumps and high voltage biasing networks. From the figure, the switched capacitance balances the IDE and CA by attaining the equilibrium by transferring the charge between the two capacitors. The switched capacitance is used to reset the voltage of the IDE capacitor on each cycle. Through simulation, we demonstrate that the proposed system can deliver satisfactory sensitivity, resolution, linearity, and power consumption, making it suitable for real-world microfluidic sensing applications. With each oscillation cycle, the switched capacitor network balances the sensing capacitance and a reference/integration capacitance, effectively sampling the sensor capacitance and encoding it as frequency. The main disadvantage is the oscillator nonlinearity. SC circuits depend on non-overlapping clock phases, switching, and careful timing which adds design overhead and may make the circuit more susceptible to clock jitter, switch timing mismatches, and noise. The main disadvantage is its gain dependency on process-voltage-temperature (PVT) variations. Authors analysed the progress of several capacitive sensor interface circuits over years and proved that the energy consumption is lowered over three times by using the capacitance to digital converters. The existing literature strongly supports the liability of CDC's that the usage of CDC improves the energy efficiency, resolution. A key enabler of this approach is the use of switched-capacitor circuits: because SC behaviour depends on capacitance ratios and a well-controlled clock frequency

3. Project Objective

The primary objective of this project is to design, implement, and analyze a frequency-domain Capacitance-to-Digital Converter using 180 nm CMOS technology. The project focuses on translating small variations in input capacitance into proportional changes in oscillation frequency using a ring oscillator-based architecture. The frequency output is further processed using a D flip-flop-based frequency divider and digital counting logic to obtain a stable and accurate digital representation of the sensed capacitance.

Additional objectives include achieving low power consumption, ensuring stable operation across process and supply variations, and obtaining a precise 50% duty-cycle digital output suitable for subsequent digital processing blocks.

3.1 Significance of the Study

The significance of this study lies in its practical approach toward designing a compact and low-power CDC suitable for integrated sensor systems. By employing frequency-domain conversion techniques, the proposed architecture minimizes the impact of analog noise and offset, which are common challenges in voltage-based CDCs. Furthermore, the use of standard CMOS building blocks such as ring oscillators and D flip-flops ensures ease of integration and scalability.

This work provides valuable insight into circuit-level design trade-offs, timing considerations, and sampling behavior in frequency-based CDCs, making it relevant for both academic learning and industrial applications.

3.2 Applications of Capacitance-to-Digital Converters

Capacitance-to-Digital Converters find applications in a wide range of domains, including but not limited to:

- Microfluidic systems for biological and chemical analysis
- Biomedical diagnostic and wearable monitoring devices
- MEMS-based pressure, acceleration, and inertial sensors
- Touchscreen and proximity sensing interfaces
- Environmental monitoring systems such as humidity and gas sensors

4. Literature Review:

Capacitive sensing is certainly very important technique in microfluidics and lab-on-chips due to its non-invasive approach, compatibility with small platforms, and ability to analyse fluids in terms of permittivity, concentration, and droplet presence. The variation in the dielectric constant of the liquid in a microchannel is a direct cause for very small changes in capacitance between the electrodes. IDE structures are the most used ones for this task, as their fringing electric fields not only provide high sensitivity to dielectric changes but also allow for the integration with CMOS technologies on the same plane, making them suitable for large-scale microfluidic sensing applications. To convert these changes in capacitance to digital signals, a variety of capacitance-to-digital converter (CDC) architectures have been introduced in the literature. SAR-based CDCs are the ones that consume less power and have a relatively simple architecture; however, their resolution and linearity are often limited by capacitor mismatch, parasitic capacitances, and charge redistribution errors, particularly in the case that they are designed for femtofarad-level resolution [1], [5]. Continuously operated sigma-delta ($\Sigma\Delta$) CDCs are able to cope with some of these drawbacks by applying oversampling and noise shaping, thus offering very high resolution and low noise levels. Nevertheless, $\Sigma\Delta$ CDCs are usually associated with complex analog circuitry that comprises of integrators and very accurate biasing networks; hence, they occupy more silicon area and consume more power thus not being appropriate for microfluidic platforms that are compact and of low power consumption [6], [16]. Incremental and zoom CDC architectures have been introduced as a way to merge the benefits of SAR and $\Sigma\Delta$ converters, thus providing higher resolution with moderate power consumption. Even though these architectures have made better trade-offs, they still mainly deal with voltage and often need very precise biasing, calibration, and large passive components, thus making the integration with microfluidic systems more complicated [22], [17]. Therefore, their use in highly integrated, low-power lab-on-chip platforms is limited.

The SC-feedback, time-encoded CDC that was originally demonstrated for MEMS capacitive microphones is not dependent on the sensing mechanism for its operating principle at all. The architecture is able to sense the capacitance directly without biasing the sensor which is exactly the case in microfluidics where the dielectric-induced capacitance variations are tiny,

slow-varying, and very much vulnerable to noise. 'Off-the-shelf' microfluidic capacitive sensing studies have focused mainly on the development of sensor designs and measurement techniques, often depending on the use of off-chip instrumentation or voltage-domain readouts which are not only limited in integration but also in scalability and power efficiency [19]. Thus, there is a significant gap in research on the application of fully integrated, high-resolution, time-domain CDC architectures to microfluidic sensing systems.

The present work is motivated by this void and extends the time-encoded CDC architecture with switched-capacitor feedback for microfluidic sensing interdigitated electrodes. The proposed method is based on the marriage of time-domain conversion, SC feedback linearization and digitalfriendly processing seeks to reach the characteristics of high sensitivity, low power consumption and robust operation which are deemed suitable for the upcoming microfluidic lab-on-chip platforms.

5. Existing CDC Architectures

Existing CDC designs include relaxation oscillators, phase-locked loop (PLL)-based converters, and ring oscillator-based systems. Among these, ring oscillator-based CDCs are attractive due to their simplicity, scalability, and ease of integration in standard CMOS processes.

5.1 Limitations of Existing Systems

Despite their advantages, conventional CDC architectures face challenges such as non-linearity, temperature sensitivity, and duty-cycle distortion in oscillator outputs. These limitations motivate the use of frequency dividers and calibration techniques to improve performance.

5.2 Summary of Research Findings

The literature survey indicates that frequency-domain CDCs provide a good trade-off between resolution, power consumption, and implementation complexity. Incorporating digital post-processing improves output stability and accuracy.

6. Research Methodology

6.1 Problem Identification

Accurate measurement of small capacitance variations is challenging due to noise, parasitic effects, and process variations. The primary problem addressed in this project is the reliable conversion of minute capacitance changes into a digital format suitable for CMOS-based systems.

6.2 System Design Approach

The proposed approach employs a frequency-domain CDC architecture. Capacitance variations modulate the oscillation frequency of a ring oscillator, which is subsequently processed using digital circuitry to obtain a digital output.

6.3 CDC Architecture and Block Description

The CDC system consists of a capacitive sensor interface, a switchedcapacitor feedback network, an OTA, a ring oscillator, a frequency divider, and a frequency-to-digital conversion block. Each block contributes to the overall conversion accuracy and stability.

6.4 Data Flow and Signal Processing

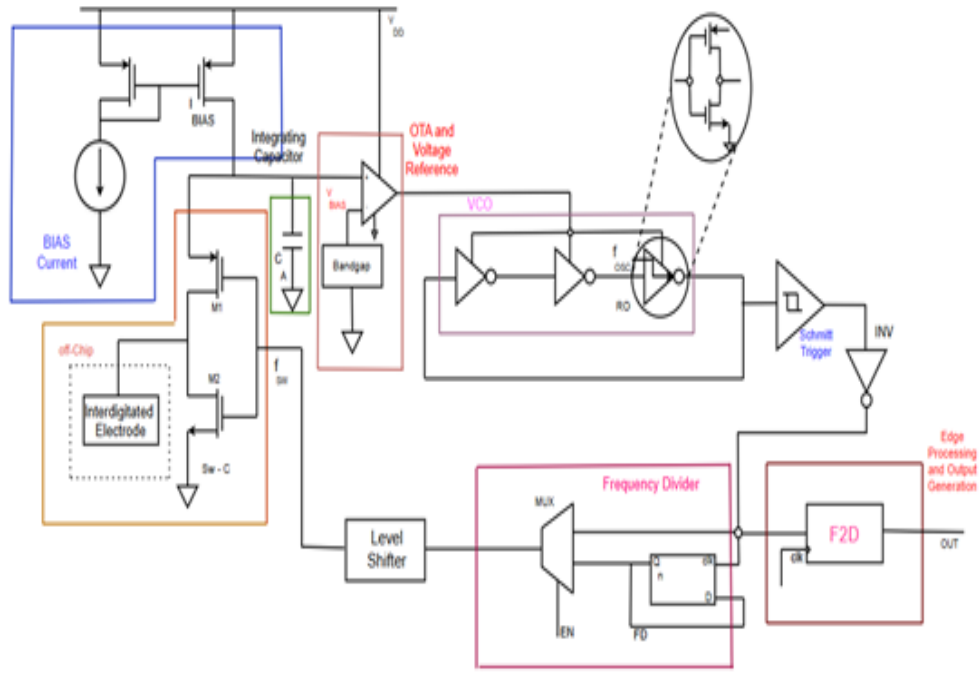
The input capacitance influences the charge transfer in the switchedcapacitor network, which generates a control voltage. This voltage modulates the OTA output current, thereby altering the ring oscillator frequency. The frequency divider and digital logic convert this frequency into a digital representation.

7. Proposed Methodology

I. Interdigitated Electrode

An IDE capacitor is used in this work to represent the sensing element. It acts as a flat capacitive transducer that is very responsive to changes in dielectric properties due to factors like humidity and the development of thin layers of water. IDE does not change capacitance by mechanical displacement and electrode separation like MEMS-based gap-varying capacitors; rather, it uses the fringe electric fields formed between coplanar electrode fingers. Therefore, the effective capacitance of an IDE largely depends on the permittivity of its surrounding medium rather than any physical movement of electrodes. Under dry ambient conditions, The IDE has a nominal rest capacitance defined by electrode geometry parameters which include finger width, spacing, overlap length, and substrate properties. When moisture or condensation is present, it increases the effective dielectric constant in the region of the fringe field; therefore, redistributes the electric field lines creating an increase in capacitance that can be measured. This allows for reliable detection of small changes in environmental dielectrics without mechanical structures. From a readout point of view, IDE is represented as a lumped capacitance inside the time encoded CDC loop. Changes in capacitance will directly modulate the oscillation frequency owing to an inverse capacitance-frequency relation imposed by switched-capacitor feedback architecture. In this work, the IDE is electrically emulated with nominal capacitance around 2.5 pF having expected variation of 100–200 fF which aligns with reported values for planar capacitive humidity and condensation sensors.

Sensing Capacitor which Represents the unknown capacitance whose variations are to be measured by the CDC. Switched-Capacitor Front End which Periodically samples the sensing capacitance and converts capacitance variations into charge packets. OTA which Integrates the transferred charge and generates a stable control voltage proportional to the sensed capacitance. Control Voltage Node Which Holds the OTA output voltage that modulates the delay of the voltage-controlled oscillator. Ring Oscillator (VCO) which Converts the control voltage into an oscillation frequency by varying inverter propagation delay. Frequency Divider which Scales down the oscillator frequency to a suitable range for digital processing



Fig(2)

II. Switched Capacitor

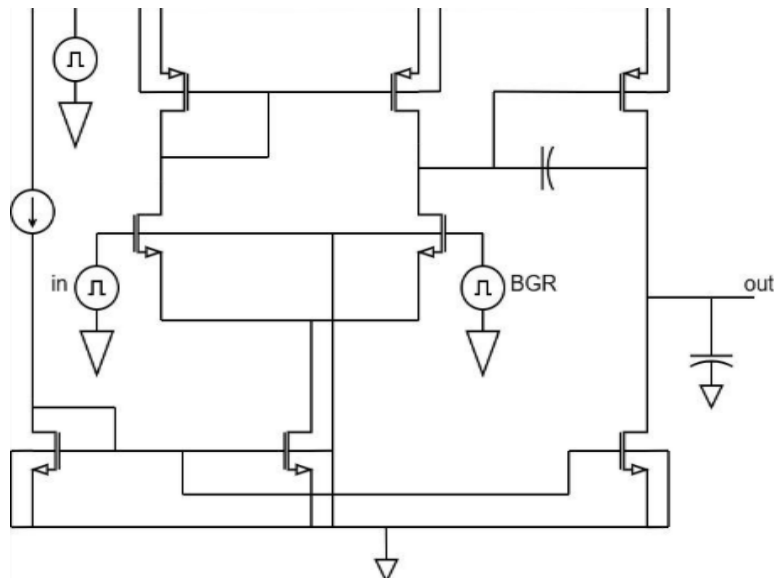
Switched-capacitor (SC) techniques are used in this work to provide a stable and scalable interface between the capacitive sensor and the time-domain readout. Instead of relying on continuous biasing elements such as high-value resistors or charge pumps, the SC network periodically transfers discrete packets of charge between capacitors using clocked switches. This enables precise control of charge flow using only capacitor ratios and clock frequency, both of which scale well with CMOS technology. In the proposed CDC, the sensed capacitance itself participates in the SC feedback loop, allowing capacitance variations to be directly translated into frequency changes while maintaining loop stability. As a result, the SC approach improves robustness against process, voltage, and temperature variations, reduces area and power overhead, and eliminates the need for external biasing circuitry, making it well suited for low-power sensor interfaces.

III. Operational Transconductance Amplifier (OTA)

An OTA means Operational Transconductance Amplifier. It is one basic building block of analog circuits. It takes a differential input voltage and gives an output current proportional to the input voltage. An OTA is different from an Operational amplifier because it does not have a voltage output and needs external feedback for stability. An OTA works as a voltage-controlled current source, so it can be used in current-mode or timebased signal processing architectures. The most important parameter for OTAs is the transconductance gain since this parameter determines how well small input voltage differences can be converted into output current. Due to high input impedance, differential operation, and compatibility with CMOS technology, OTAs are widely used in integrated sensor interfaces. In a capacitance-to-digital converter, the operational transconductance amplifier is the key front-end analog interface between the capacitive sensor and the downstream digital conversion stage based on time or frequency. The primary function of an OTA in CDC is to amplify small voltage swings due to femtofarad-level capacitance changes at the sensor interface and linearize them, converting these into proportional currents. If this currentmode operation could be directly integrated onto a capacitor, then precise voltages would be generated which could drive a voltage-controlled oscillator. High input impedance from the OTA means that sensitive capacitive sensors will not load down by any following circuitry necessary for measurement accuracy. In addition to this common mode disturbances like parasitic capacitances temperature drift and supply noise are suppressed because of OTAs differential nature. In architectures for humidity sensing and biofluid CDC where signals vary slowly with high noise sensitivity, OTAs improve resolution as well as signal-to-noise ratios providing longterm stability. The VCO control voltage rate change determines its output frequency; hence digital resolution of the CDC is directly affected.

The OTA provides a linear and continuous mapping from capacitance variations to frequency variations. Its transconductance g_m has a direct impact on the overall gain of the CDC and therefore affects the resolution and sensitivity. A higher g_m increases the current for a given variation of voltage; this results in a faster voltage ramp across CA and then a larger frequency shift at the VCO.

$$I_{out} = g_m(V_+ - V_-)$$



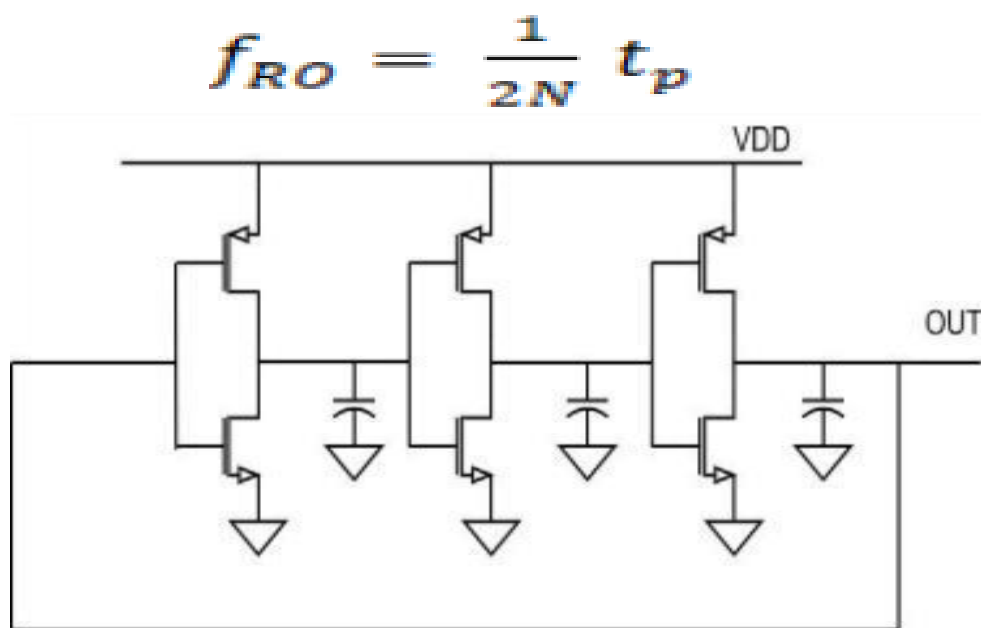
V. Bandgap reference circuit

The BGR is a bandgap reference that gives a temperature independent reference voltage used to bias the OTA. This voltage reference is made from the combination of a CTAT junction voltage and PTAT produced by matched exponential devices operating at different current densities. By scaling and summing these two components, first order temperature dependence can be cancelled out, thus resulting in a stable reference voltage over an operating temperature range. The equation for the reference voltage then becomes are the scaling resistors. The resistor ratio is chosen so that the temperature coefficients of CTAT and PTAT components cancel each other out, giving a reference voltage with very small variation with temperature. This stable reference voltage will be used to bias the OTA so that there will be consistent transconductance and good analog performance over temperature.

$$V_{ref} = V_{CTAT} + \alpha V_{PTAT}$$

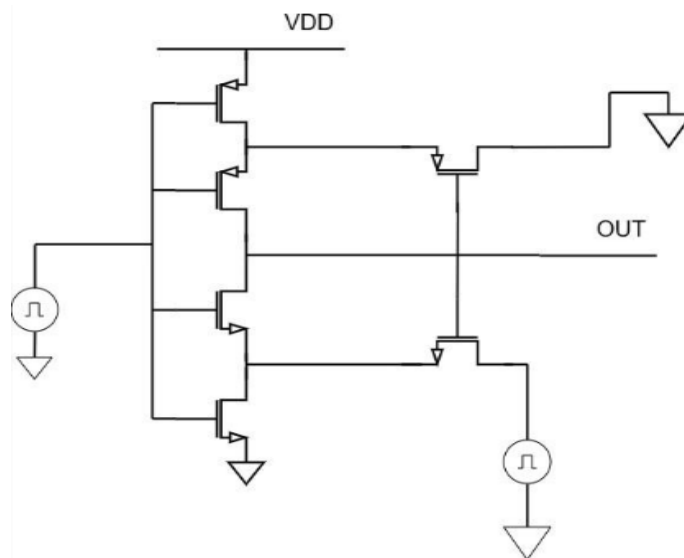
V. Ring Oscillator:

It is a simple type of oscillator created by chaining an odd number of inverters in a loop fashion, in which the output keeps on toggling because of the inherent propagation delay introduced by each stage. It requires an inverse proportional number of stages and their delays for its oscillation frequency. Ring oscillators find various applications in VCOs, CDCs, and on-chip clocks due to their compact nature, ease of integration, and simplicity, but they are prone to noise, temperature, and processing variations, which can influence the frequency. In this design, the ring oscillator is used as the readout element of the CDC rather than a separate ADC, which keeps the interface simple. The OTA-based switched-capacitor front end converts the unknown capacitance into a control voltage that directly affects the inverter delay of the oscillator. As the delay changes, the oscillation frequency shifts accordingly while the oscillator runs continuously. The resulting frequency is divided and counted over a fixed time window, providing a digital output that tracks variations in the sensed capacitance.



VI. Schmitt Trigger

A Schmitt trigger is used at the output of the ring oscillator in the proposed time-encoded capacitance-to-digital converter (CDC) for robust digital signal regeneration. If directly processed by digital blocks, since the oscillator output may have slow transitions and noise jitter due to analog front-end impairments, this will create timing uncertainty; therefore, it is critical to avoid spurious switching due to noise and jitter by introducing intentional hysteresis through separate rising and falling thresholds defined by the Schmitt trigger. This hysteresis not only improves edge stability but also ensures that clocks are generated consistently under various operating conditions. The rail-to-rail digital waveform output then gets buffered and sent to both the frequency divider as well as the frequency-to-digital (F2D) block. The Schmitt trigger helps improve timing robustness by decoupling the mixed-signal oscillator from the digital backend without changing oscillation frequency; hence, it maintains time-encoded CDC operation fidelity.



VII. Frequency Divider

A master–slave D flip-flop–based divide-by-2 frequency divider is implemented in 180-nm CMOS technology. The master–slave topology, formed by two cascaded level-sensitive latches driven by complementary clock phases, ensures edge-triggered operation and eliminates race-through. By feeding the complementary output back to the data input, the flip-flop toggles its state at every active clock edge, resulting in an output frequency equal to half of the input clock frequency. This structure provides a stable

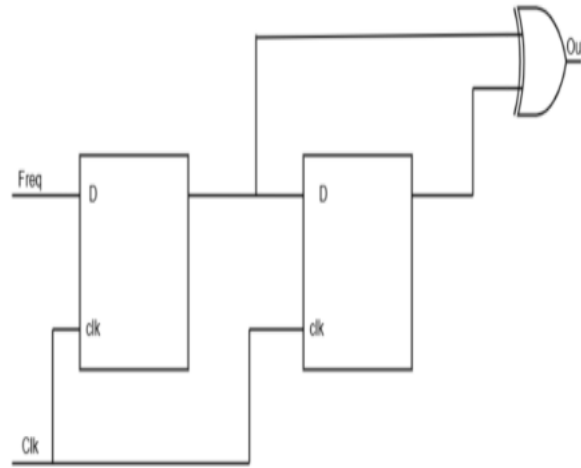
50% duty cycle, full-swing digital output, and reliable operation over a wide frequency range. In the proposed capacitance-to-digital converter, the sensed capacitance modulates the oscillation frequency of a ring oscillator. Since this frequency is typically higher than what the digital processing blocks can directly handle, the divide-by-2 frequency divider performs frequency scaling to ensure compatibility with the subsequent frequency-to-digital conversion stage. Moreover, the divider delivers a symmetrical duty cycle clock that simplifies digital counting and improves robustness against jitter and high-frequency noise components present at the oscillator output. Consequently, the frequency divider serves as a critical interface between the analog sensing front-end and the digital back end, enabling stable and accurate capacitance measurement.

VIII. Frequency-to-Digital Converter for Microfluidic

Capacitance Sensing

The Frequency-to-Digital Converter (F2D) in the suggested microfluidic capacitance-to-digital converter (CDC) is in charge of converting the timeencoded oscillator output into a digital representation of the detected capacitance. The F2D serves as the main interface between the analog sensing front-end and the digital processing domain because changes in the

interdigitated electrode (IDE) capacitance directly modulate the oscillation frequency. The F2D is naturally suited for low-frequency, high-resolution microfluidic sensing applications because, in contrast to voltage-domain ADCs, it functions solely in the time domain.



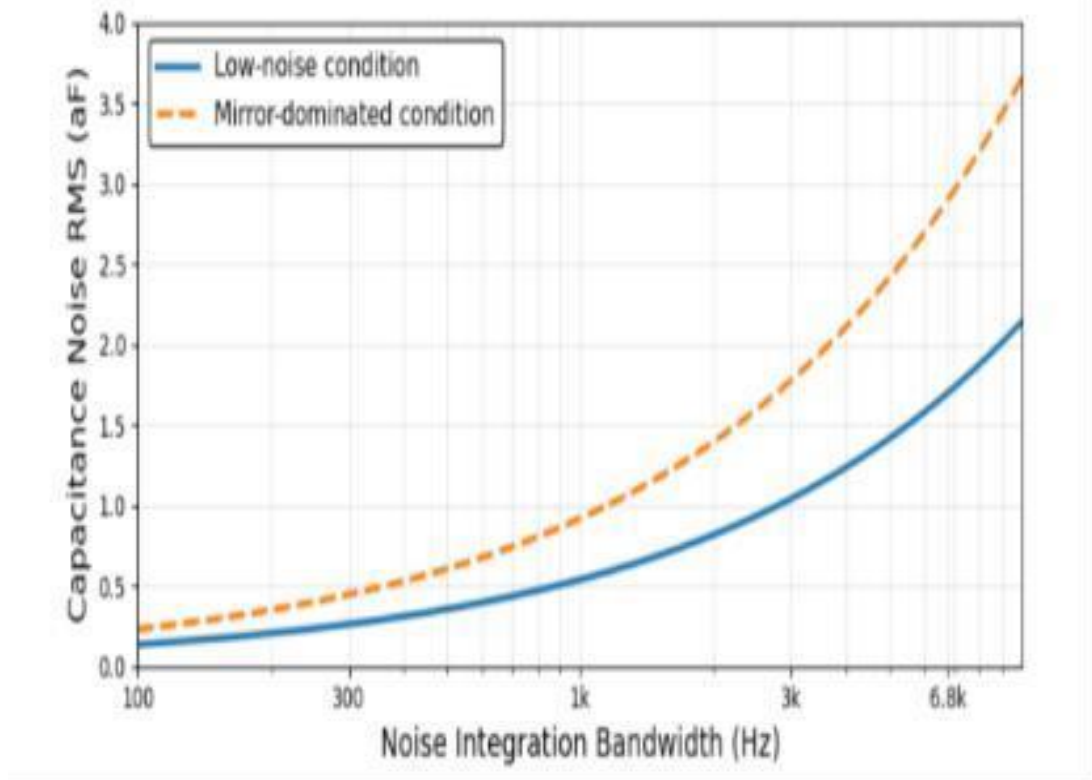
The F2D block accepts a frequency-modulated input signal and a reference clock (*clk*) as its inputs. The input signal carries the sensed information encoded in its frequency, while the clock defines the observation window for conversion. The block produces a digital output (*OUT*) whose value is proportional to the input frequency, thereby providing a digital representation of the time-encoded signal suitable for subsequent digital processing. The output of the capacitance-controlled ring oscillator is a frequency-modulated digital signal whose instantaneous frequency varies inversely with the sensor capacitance.

8. Results and Discussion

I. Noise Performance and Capacitance Resolution

The CDC's output noise was analysed over the 300 Hz–6.8 kHz, following the convention used in high-resolution microfluidic CDC systems. The noise behaviour of the CDC was evaluated under two different OTA operating conditions. In the first operating point, the circuit shows an integrated noise of $1.221 \times 10^{-12} \text{ V}^2$, which corresponds to about $1.105 \mu\text{V}$ RMS at the output. This level of voltage noise translates to a capacitance

resolution of roughly 2.90aF. When the OTA is biased more favourably, the overall noise reduces noticeably. Under this improved condition, the integrated noise drops to $4.372 \times 10^{-13} \text{ V}^2$, giving a voltage noise of 0.661 μV RMS and a finer capacitance resolution of 1.72aF. Taken together, these measurements show that the proposed CDC consistently operates within a 1.7–2.9aF resolution window, which is significantly



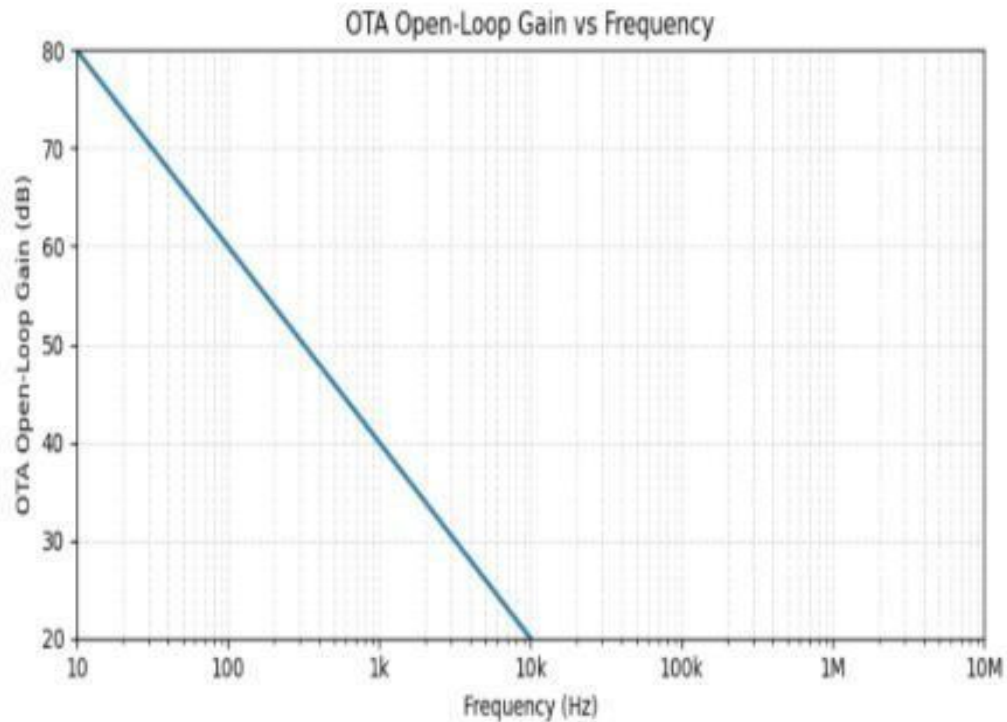
Fig(g)

better than what is typically achieved in standard CMOS CDC designs. The overall recommended noise range of the proposed CDC is therefore 1.7–2.9aF, which is significantly better as “A capacitance resolution of 6.3aF is achieved within a 1-ms conversion interval” [12]

II. OTA V^+ Node Stability:

The OTA non-inverting node is kept at a voltage level of 1.4 V. This is a stable voltage. The 15-pF integration capacitor helps to reduce the switching-induced ripple a lot. The remaining variation at V^+ is mainly because of the OTA input-referred noise. This noise is around 0.661–1.105

$\mu\text{V RMS}$. So, the deterministic ripple is low it is below the noise floor. This voltage noise is like a capacitance noise of 1.72–2.90 aF through the OTA loop gain. This helps to reduce the bias-induced nonlinearity and the sensitivity, to OTA switching artifacts. The OTA non-inverting node and the OTA loop gain work together to make this happen. Therefore, oscillator frequency variations are mainly due to capacitance changes rather than bias fluctuations, improving repeatability and robustness that is very important in microfluidic sensing where effective dielectrics can vary dynamically



Fig(h)

III. Contribution of SC Feedback to Parasitic:

The new system stands apart from traditional designs that maintain a constant charge. Conventional systems often rely on components like resistors and charge pumps, which can introduce complications and reduce reliability, especially under extreme temperatures. While the new system brings improvements, there are still some instances where it can be unreliable. The new system uses a technique called switched-capacitor feedback to stabilize the VCO loop and improve performance. When microfluidic samples go in or out of the sensing area the oscillator frequency changes. This change happens in proportion to the inverse of the capacitance of the samples. The system just keeps on oscillating without any drift or loss of lock. The microfluidic samples can be detected in time without any problems even when the dielectric changes of the microfluidic samples are happening really fast. The design

of the system is better because it does not need biasing components, like charge pumps and high-value resistors.

9. Comparing With Existing CDC Technology

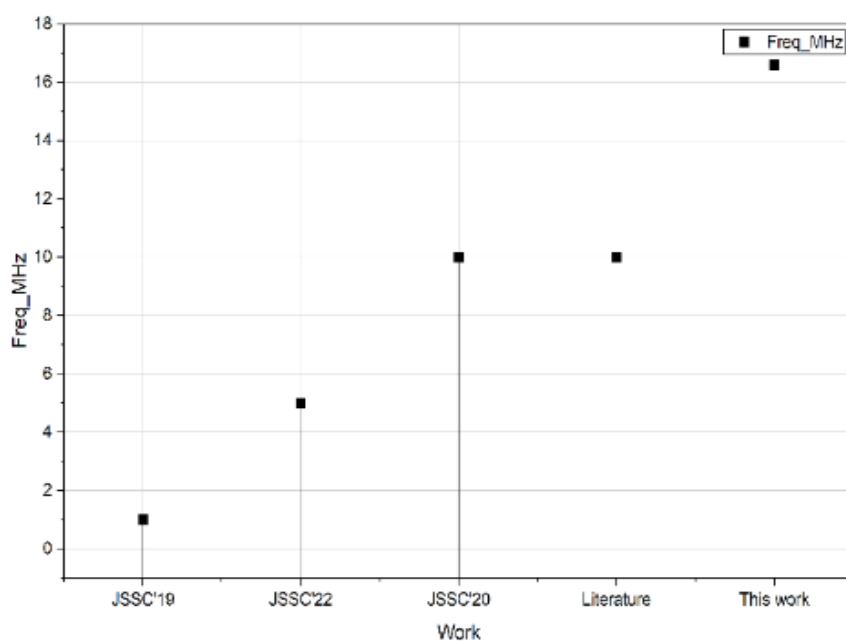
Parameter	SAR CDC[13] (JSSC'19)	CT- $\Sigma\Delta$ CDC[16] (JSSC'22)	Incremental / Zoom CDC[17] (JSSC'20)	Microfluidic CDC (Literature)	This Work
CMOS Technology	65 nm	180nm	55 nm	180–350 nm	180 nm
CDC Architectur e	SAR- based	Continuous -Time $\Sigma\Delta$	Incremental Zoom	Voltage/ Time- based	TimeDomain CDC
Conversion Domain	Charge / Voltage	Voltage ($\Sigma\Delta$)	Voltage / Time	Voltage / Time	Time / Frequency

Sensor Type	Generic capacitive	Generic capacitive	Generic capacitive	IDE/Coplane r electrodes	IDE (microfluidic channel)
-------------	-----------------------	-----------------------	-----------------------	-----------------------------	----------------------------------

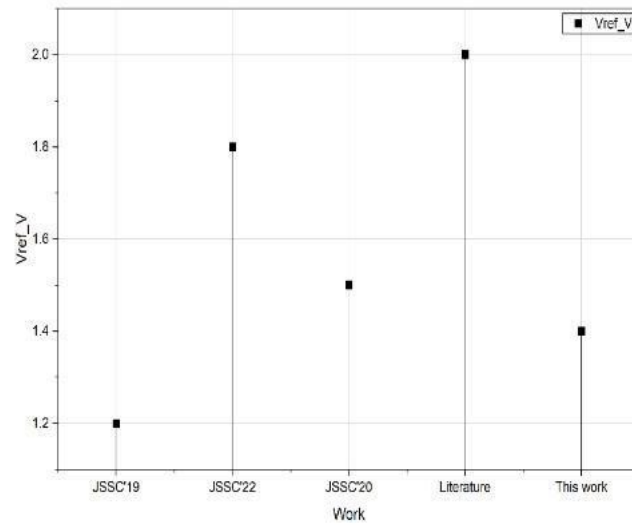
Target Application	Generic sensing	Precision sensing	Generic sensing	Microfluidic dielectric sensing	Microfluidic analyte sensing
Rest Capacitance	1–10 pF	5–50 pF	2–20 pF	1–5 pF	2.575 pF
Bias Current, I_{BIAS}	<10 μA	50–200 μA	20–100 μA	10–100 μA	39 μA
Reference Voltage, V_{BIAS}	0.8–1.2 V	1.0–1.8 V	1.0–1.5 V	1.0–2.0 V	1.4 V
Oscillation / Sampling Frequency	<1 MHz	<5MHz	1–10 MHz	<10MHz	16.6 MHz
Frequency Divider Ratio, NN	----	---	1-4	1-4	2
Noise Integration Bandwidth	≤ 10 kHz	≤ 10 kHz	≤ 5 kHz	≤ 1 kHz	300 Hz – 6.8 kHz

Capacitance Noise (RMS)	~10 aF	21–115 aF	8–15 aF	10–100 aF	1.7 – 2.9 aF
Sensor Biasing Required	Yes	Yes	Yes	Yes	No

Oscillation Frequency



Operating Voltage



10. Conclusion

This work introduces a **closed-loop readout architecture for capacitive sensors** designed to detect variations in physical gaps. Traditional

capacitive sensing systems often require dedicated sensor biasing circuitry to achieve accurate measurements of small capacitance changes, which increases circuit complexity, area, and power consumption. The proposed architecture overcomes these limitations by embedding the sensing capacitor directly within a feedback loop. This configuration eliminates the need for separate biasing networks and allows for **direct capacitance-to-frequency conversion**, reducing the analog overhead that is typically required in conventional designs.

The feedback loop enables the system to translate minute changes in capacitance into corresponding variations in output frequency, simplifying the signal processing chain. This approach is particularly suitable for **compact and low-power implementations**, making it well-suited for modern sensing applications, including portable or wearable devices. In the experimental phase, a voltage-controlled capacitance, realized with a varactor diode and an external biasing network, was used to emulate the sensor input. This setup provides a controlled environment for testing and validation in the absence of a physical capacitive sensor, ensuring repeatable and reliable measurements.

Experimental results confirm the feasibility of the proposed concept, demonstrating a peak **signal-to-noise-and-distortion ratio (SNDR)** of approximately 50 dB over a measurement bandwidth ranging from 300 Hz to 6.8 kHz. These results highlight that while the architecture effectively converts capacitance variations into frequency signals, the current prototype is constrained by noise contributions from the front-end circuitry. This limitation indicates that further optimization is necessary to fully exploit the potential of the design.

Future improvements could include **enhanced noise-shaping techniques**, which suppress unwanted noise and improve resolution, a **fully differential implementation** to mitigate common-mode interference, and careful circuit optimization to **reduce power consumption**. With these enhancements, the architecture has the potential to achieve higher resolution and broader applicability in precision sensing applications.

PO & PSO Attainment

PO. No	Graduate Attribute	Attained	Justification
PO 1	Engineering knowledge	Yes	Working with electronical machines
PO 2	Problem analysis	Yes	Development needed to build a product
PO. No	Graduate Attribute	Attained	Justification
PO 3	Design/Development of solutions	Yes	Reduce the more complex design into simple
PO 4	Conduct investigations of complex problems	Yes	Conversations with mentors about the problems
PO 5	Modern Tool usage	Yes	Robotic arms
PO 6	The Engineer and society	Yes	It provides well qualified products
PO 7	Environment and Sustainability	Yes	It is an eco friendly making process
PO 8	Ethics	Yes	Long time use with less cost
PO 9	Individual and team work	Yes	Team work makes the work easier

PO 10	Communication	Yes	It is used to learn and understand things effectively
PO 11	Project management and finance	No	
PO 12	Life-long learning	Yes	Learning about the industry environment

PSO.No	Graduate Attribute	Attained	Justification
PSO 1	To analyze, design and develop solutions by applying the concepts of Robotics for societal and industrial needs.	Yes	Robotic arms were used in major production areas, which makes the work faster, easier
PSO 2	To create innovative ideas and solutions for real time problems in Manufacturing sector by adapting the automation tools and technologies.	Yes	By using some of the softwares , we can make creative designs, which helps to pre verify the problems in any fields

